

Fourth-Order Chronological Carry: Connected Quadruples, Extension Parameters, and the Stirling Shadow

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Abstract

A third-order chronological carry refinement (CCR-3) assigns arithmetic weights to singleton, adjacent-pair, and consecutive-triple interactions on an ordered path of correction events. The third-order recurrence is rigid once those three weights are chosen, but it does not determine what must happen at order four. This paper isolates the exact obstruction and gives a fourth-order extension theory.

On a path chronology there is exactly one new connected support type of cardinality four: a consecutive quadruple. Algebraically, the first three CCR bands already generate every degree-four contribution obtained by appending a terminal block of size one, two, or three. A genuine fourth-order refinement therefore requires one additional coefficient multiplying $u^4 P_{n-4}$. We prove that CCR-3 and chronological locality alone cannot determine that coefficient.

We then impose a precisely typed fourth-order polarization: compatibility with CCR-3, path-locality, stationarity after falling-factorial normalization, and p -divisibility of the new connected source. For prime $p \geq 5$, every such extension is classified by one integer parameter λ :

$$\begin{aligned} P_{p,n}^{[4,\lambda]}(u) = & (1 + \ell_n u) P_{p,n-1}^{[4,\lambda]}(u) \\ & + p(n-1) \ell_{n-1} u^2 P_{p,n-2}^{[4,\lambda]}(u) \\ & - p(n-2)(n-1) u^3 P_{p,n-3}^{[4,\lambda]}(u) \\ & + \lambda p(n-3)(n-2)(n-1) u^4 P_{p,n-4}^{[4,\lambda]}(u) \pmod{u^5}, \end{aligned}$$

where $\ell_n = (p+1)n - p$. The parameter is realized by a fourth lower-Hessenberg band and by the operator

$$\mathcal{H}_{p,\lambda} = D + L_p - pXL_p - pX^2 - \lambda pX^3.$$

For an arbitrary p -divisible fourth source $q_{p,m} = p(m-1)\underline{3}\eta_m$, falling-factorial normalization gives an exact inverse law. If Δ_m is the difference between its fourth normalized column and the zero-source column, then

$$\eta_m = \frac{m\Delta_{m+1} - (m-4)\Delta_m}{p}.$$

Thus the full connected-source sequence is reconstructible from consecutive fourth-column rows. Stationary sources are exactly those for which Δ_m is constant, and in the one-parameter family this specializes to

$$\Pi_{p,4}^{(\lambda)}(m) = \Pi_{p,4}^{(0)}(m) + \frac{\lambda p}{4}.$$

The first three columns are completely insensitive to the fourth source.

We also give the explicit quartic formula for $\Pi_{p,4}^{(\lambda)}$ and prove that every member of the family has the same fourth-jet Stirling shadow

$$P_{p,n}^{[4,\lambda]}(u) \equiv \prod_{j=1}^n (1 + ju) \pmod{(p, u^5)}.$$

Thus the mod- p shadow erases the new connected parameter even though the exact fourth column recovers it. At $p = 5$ the raw coefficient congruence remains valid but the normalized fourth shadow encounters a denominator resonance because 5760 is not invertible modulo 5. The contribution is family-specific: cumulant/Möbius decompositions, Stirling numbers, and banded Hessenberg recurrences are classical tools and are not claimed as new.

Keywords. chronological carry refinement; deterministic pushdown automata; connected interactions; fourth-order recurrence; Hessenberg matrix; Stirling numbers of the first kind; fixed-column transport; identifiability.

1 Introduction

The arithmetic refinement studied here is rooted in one local operation of a completion-aware deterministic pushdown automaton. In the original 2:1 machine shown in Figure 1, the heavier loop in the left state is the distinguished online cancellation $0, 0/\varepsilon$. Reading that cancellation through the associated path model produces an ordered list of correction events. The realization cost of the resulting correction signatures is nearest-neighbor in chronological order, so the event positions carry a natural path graph

$$1 - 2 - \cdots - n.$$

That path-local chronology is the structural input to the carry refinement.

The scientific boundary is essential. The DPDA-derived branch supplies (i) a distinguished correction event, (ii) chronological ordering, and (iii) nearest-neighbor locality on the path of correction positions. The numerical weights assigned to connected supports are an additional arithmetic polarization. In particular, the third-order recurrence recalled in Section 2 is a chosen arithmetic lift of the path-local skeleton, not a theorem that the machine transition table uniquely determines those coefficients.

The third-order model, CCR-3, uses the only connected path supports of sizes one, two, and three: a singleton, an adjacent pair, and a consecutive triple. At order four, the geometry remains simple but the arithmetic changes qualitatively. There is again only one new connected support type, namely a consecutive quadruple, yet the coefficient assigned to it is not recoverable from the third-order data. This is the obstruction addressed here.

Main results. The paper proves the following.

- (i) A four-event connected chronological support on a path is necessarily a consecutive quadruple. The degree-four contributions generated by terminal blocks of sizes at most three are already determined by CCR-3; a terminal size-four block is the sole new datum.
- (ii) CCR-3 plus chronological locality does not determine a numerical CCR-4 coefficient. For an arbitrary p -divisible connected source, consecutive fourth-column rows recover the entire normalized source sequence by an exact inverse transport formula. Stationarity is equivalent to a constant fourth-column translation, and the stationary extensions form a one-parameter family indexed by $\lambda \in \mathbb{Z}$.
- (iii) The λ -family has an exact four-band lower-Hessenberg realization and the operator form

$$\mathcal{H}_{p,\lambda} = D + L_p - pXL_p - pX^2 - \lambda pX^3.$$

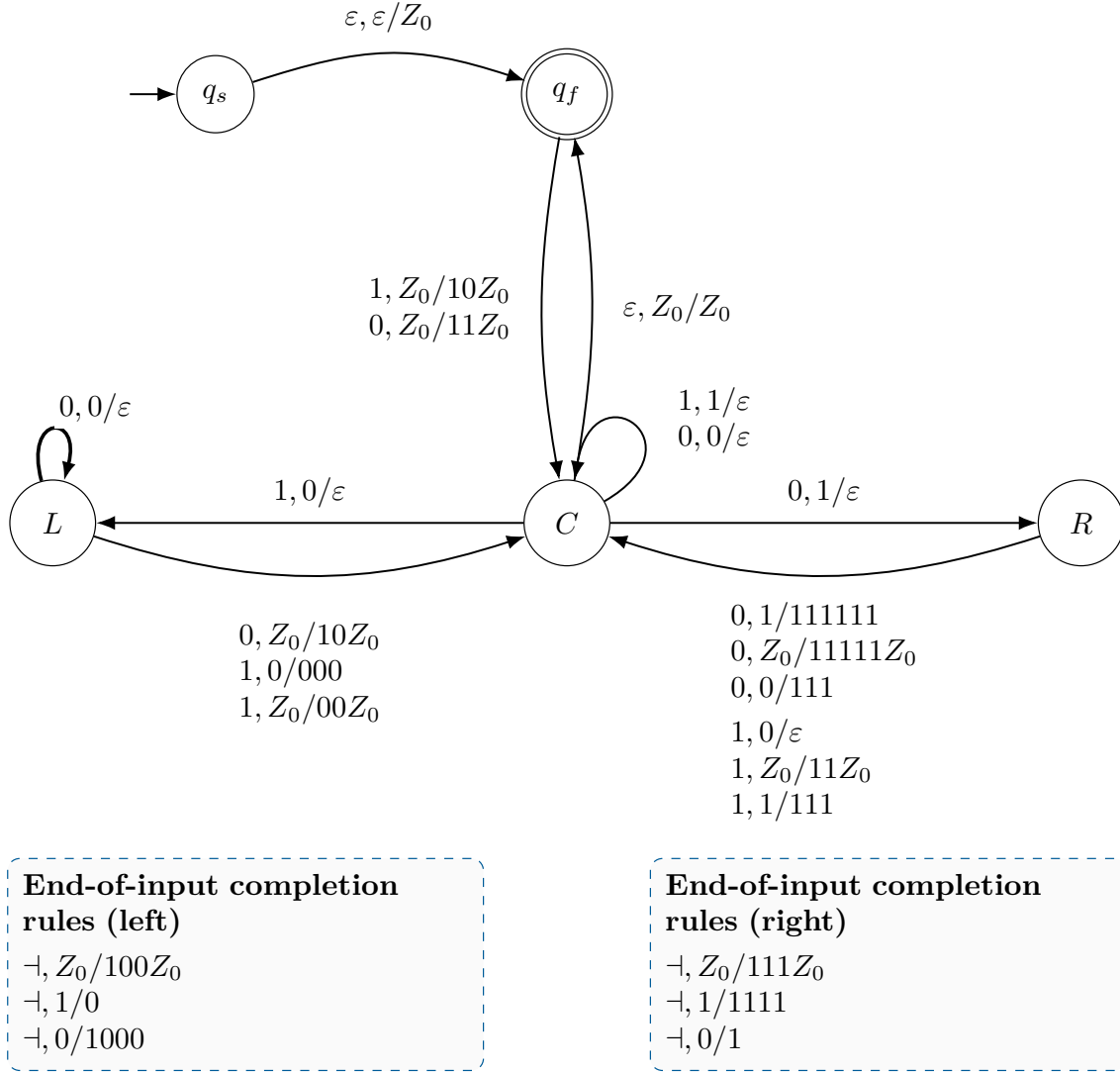


Figure 1: The motivating completion-aware 2:1 DPDA. The heavier left-state loop is the distinguished cancellation from which the chronological correction events are extracted. Solid arcs process binary input; the blue dashed boxes preserve the right-endmarker completion lineage. All textual labels are black. The figure is provenance and motivation only: the arithmetic coefficients below are not claimed to be forced uniquely by this transition table.

A primitive same-sign band convention selects $\lambda = 1$ as one concrete CCR-4 polarization, but this selection is not attributed to the DPDA.

- (iv) The normalized fourth column obeys a first-order transport law. In the general p -divisible class it exactly identifies the connected-source sequence; in the stationary subfamily, changing the fourth polarization translates the entire normalized fourth column by exactly $p(\lambda - \mu)/4$.
- (v) The fourth column is an explicit quartic polynomial. Every λ has the same mod- p fourth-jet Stirling shadow, so the free shadow cannot see the connected quadruple parameter even though the exact fourth column identifies it from one row.
- (vi) The raw fourth-order Stirling congruence is valid also at $p = 5$, but the normalized polynomial shadow cannot be reduced by division there because its natural denominator 5760 is divisible by 5.

Novelty boundary. Moment–cumulant relations as Möbius inversion on partition lattices are classical [5, 4], as are carry matrices [3], Stirling numbers [2], and determinant recurrences arising from banded lower-Hessenberg matrices [1]. We use these as context and tools. The contribution claimed here is narrower: the exact fourth-order extension and identifiability structure of this chronological carry family. A targeted literature search found adjacent theories but no direct prior theorem matching the one-parameter CCR-3-compatible fourth-order family, its constant fourth-column translation law, and its common fourth-jet Stirling shadow.

2 The CCR-3 baseline

Fix a prime $p \geq 5$ and define

$$\ell_n = (p+1)n - p.$$

The third-order arithmetic polarization is taken as the baseline object of this paper.

Definition 2.1 (CCR-3 baseline). Work in $\mathbb{Z}[u]/(u^4)$ and set

$$P_{p,0}^{[3]}(u) = 1, \quad P_{p,-1}^{[3]}(u) = P_{p,-2}^{[3]}(u) = 0.$$

For $n \geq 1$, define

$$\begin{aligned} P_{p,n}^{[3]}(u) = & (1 + \ell_n u) P_{p,n-1}^{[3]}(u) + p(n-1) \ell_{n-1} u^2 P_{p,n-2}^{[3]}(u) \\ & - p(n-2)(n-1) u^3 P_{p,n-3}^{[3]}(u) \pmod{u^4}. \end{aligned} \quad (1)$$

The three sources are called the singleton slot, adjacent-pair interaction, and connected triple interaction.

For $0 \leq k \leq 3$ put

$$C_{n,k}^{(p)} = [u^k] P_{p,n}^{[3]}(u), \quad \Pi_{p,k}(m) = \frac{C_{m-1,k}^{(p)}}{(m-1)_{\underline{k}}} \quad (m \geq k+1).$$

Coefficient extraction from (1) gives the transport equations

$$\begin{aligned} m \Pi_{p,1}(m+1) &= (m-1) \Pi_{p,1}(m) + \ell_m, \\ m \Pi_{p,2}(m+1) &= (m-2) \Pi_{p,2}(m) + \ell_m \Pi_{p,1}(m) + p \ell_{m-1}, \\ m \Pi_{p,3}(m+1) &= (m-3) \Pi_{p,3}(m) + \ell_m \Pi_{p,2}(m) \\ &\quad + p \ell_{m-1} \Pi_{p,1}(m) - p. \end{aligned} \quad (2)$$

Proposition 2.2 (First three normalized columns). *For the indicated ranges,*

$$\Pi_{p,1}(m) = \frac{(p+1)m - 2p}{2},$$

$$\Pi_{p,2}(m) = \frac{p+1}{24} \left[3(p+1)m^2 - (5p+1)m - 12p \right],$$

and

$$\Pi_{p,3}(m) = \frac{p+1}{48} F_p(m),$$

where

$$F_p(m) = (p+1)^2 m^3 + (p^2 - 1)m^2 - (34p^2 + 18p)m + 56p^2 - 8p.$$

The boundary values are

$$\Pi_{p,1}(2) = 1, \quad \Pi_{p,2}(3) = p+1, \quad \Pi_{p,3}(4) = (p+1)^2.$$

Proof. Substitution into (2) verifies the three displayed polynomials. Each transport equation is first order in its current column, so the corresponding boundary value gives uniqueness. No computer-algebra interpolation is used as a proof premise. $\square$

Modulo p , every non-singleton source in (1) disappears and $\ell_n \equiv n$. Hence

$$P_{p,n}^{[3]}(u) \equiv \prod_{j=1}^n (1 + ju) \pmod{(p, u^4)}.$$

The purpose of T8 is to determine exactly what survives, and what becomes new, when the truncation is raised to u^5 .

3 What is genuinely new at order four

Definition 3.1 (Connected chronological support). A set of correction positions is *connected* if it induces a connected subgraph of the chronological path $1 - 2 - \dots - n$.

Lemma 3.2 (Unique connected quadruple). *Every connected chronological support of cardinality four is an interval of the form*

$$\{j, j+1, j+2, j+3\}.$$

Thus there is exactly one new connected support type at order four.

Proof. Every connected subset of a path graph is an interval. An interval with four vertices has the displayed form. $\square$

The structural issue is not the shape of the new support but the arithmetic weight assigned to it. The recurrence itself gives a clean way to separate propagated lower-order terms from the genuinely new fourth-order source.

Definition 3.3 (Terminal block form through order four). A path-local fourth-order extension of CCR-3 is a recurrence in $\mathbb{Z}[u]/(u^5)$ of the form

$$\begin{aligned} P_{p,n}(u) = & (1 + \ell_n u)P_{p,n-1}(u) + p(n-1)\ell_{n-1}u^2P_{p,n-2}(u) \\ & - p(n-2)(n-1)u^3P_{p,n-3}(u) + q_{p,n}u^4P_{p,n-4}(u) \pmod{u^5}, \end{aligned} \quad (3)$$

with $P_{p,0} = 1$ and $P_{p,n} = 0$ for negative n . The first three terminal block weights are fixed by CCR-3; $q_{p,n}$ is the connected quadruple source.

Proposition 3.4 (Lower-order block separation). *In (3), every contribution to $[u^4]P_{p,n}$ obtained by appending a terminal connected block of size one, two, or three is already determined by CCR-3. The only term in the fourth coefficient not generated by those lower bands is $q_{p,n}$ itself.*

Proof. Extracting u^4 from (3) gives

$$\begin{aligned} [u^4]P_{p,n} = & [u^4]P_{p,n-1} + \ell_n[u^3]P_{p,n-1} \\ & + p(n-1)\ell_{n-1}[u^2]P_{p,n-2} \\ & - p(n-2)(n-1)[u]P_{p,n-3} + q_{p,n}. \end{aligned} \quad (4)$$

The middle three lines attach a terminal block of size one, two, or three to a previously generated lower-order history. Iterating those terms produces exactly the degree-four histories decomposable into lower connected block sizes, including the patterns $1 + 1 + 1 + 1$, $2 + 1 + 1$, $2 + 2$, and $3 + 1$ in chronological order. The final scalar $q_{p,n}$ is the contribution of a single terminal block of size four. By Theorem 3.2, that block is the unique consecutive quadruple support. $\square$

Remark 3.5 (Relation with cumulant language). Classical cumulants isolate connected information by Möbius inversion on the lattice of set partitions [5, 4]; mixed cumulants vanish when the variables split into independent families. That theory motivates the word “connected,” but Theorem 3.4 does not identify the CCR coefficients with probabilistic cumulants. It is an internal recurrence statement about terminal interval blocks on a path.

Theorem 3.6 (Fourth-order non-identifiability from CCR-3). *CCR-3 and chronological path-locality do not determine a unique fourth-order numerical weight. More precisely, for any integer sequence $(q_{p,n})_{n \geq 4}$, recurrence (3) has exactly the same coefficients through u^3 as CCR-3.*

Proof. The new term is divisible by u^4 , so it cannot affect any coefficient of degree at most three. Induction on n therefore shows that every choice of $q_{p,n}$ has the same third jet as (1). The fourth coefficient is the first place where the choices can differ. $\square$

This theorem is the central scope correction at order four: a numerical CCR-4 coefficient must come from an additional arithmetic polarization, not from third-order data alone.

3.1 Exact reconstruction of a general p -divisible fourth source

Suppose the connected source is p -divisible after the natural size-four placement factor:

$$q_{p,m} = p(m-1)\underline{3}\eta_m, \quad \eta_m \in \mathbb{Z}, \quad m \geq 4.$$

For the corresponding solution of (3), write

$$C_{n,k}^{(q)} = [u^k]P_{p,n}(u), \quad \Pi_{p,4}^{(q)}(m) = \frac{C_{m-1,4}^{(q)}}{(m-1)\underline{4}}, \quad m \geq 5.$$

Let $\Pi_{p,4}^{(0)}$ denote the zero-fourth-source solution and put

$$\Delta_m = \Pi_{p,4}^{(q)}(m) - \Pi_{p,4}^{(0)}(m).$$

Theorem 3.7 (Fourth-source reconstruction). *For every $m \geq 5$,*

$$\boxed{m\Delta_{m+1} = (m-4)\Delta_m + p\eta_m.} \tag{5}$$

At the boundary,

$$\boxed{\Delta_5 = \frac{p\eta_4}{4}.} \tag{6}$$

Consequently the connected source is exactly identifiable from the fourth normalized column:

$$\boxed{\eta_m = \frac{m\Delta_{m+1} - (m-4)\Delta_m}{p} \quad (m \geq 5),} \tag{7}$$

with $\eta_4 = 4\Delta_5/p$.

Proof. Extract u^4 from (3), put $n = m$, and divide by $(m)\underline{4} = m(m-1)(m-2)(m-3)$. The lower three source bands are identical for the given source and for the zero-source recurrence because all columns of degree at most three agree by Theorem 3.6. Their difference therefore cancels. The remaining fourth source contributes

$$\frac{q_{p,m}}{(m)\underline{4}} = \frac{p(m-1)\underline{3}\eta_m}{(m)\underline{4}} = \frac{p\eta_m}{m},$$

while the preceding fourth coefficient contributes $(m-4)\Delta_m/m$. Multiplying by m gives (5). At $m = 4$, the only difference in $C_{4,4}$ is $q_{p,4} = 6p\eta_4$; division by $(4)\underline{4} = 24$ gives (6). Solving (5) for η_m yields (7). $\square$

Thus fourth-order identifiability is not restricted to the stationary model: the entire p -divisible connected chronology can be read back from the fourth column. Stationarity becomes a testable property of that column rather than merely an imposed coefficient ansatz.

4 The stationary p -divisible fourth-order family

We now specify one natural class in which the new fourth-order datum can be classified exactly.

Definition 4.1 (Stationary p -divisible fourth polarization). A recurrence of the form (3) is *stationary p -divisible* if, for every $n \geq 4$,

$$p(n-1)_3 \mid q_{p,n},$$

and the normalized integer quotient

$$\eta_n = \frac{q_{p,n}}{p(n-1)_3}$$

is independent of n . Thus stationarity is imposed only after removing the inherited factor p and the falling-factorial normalization of a size-four terminal block.

Theorem 4.2 (Stationarity and one-parameter classification). *For every prime $p \geq 5$ and every p -divisible fourth-order extension, the following are equivalent:*

- (i) *the normalized connected source η_n is independent of n ;*
- (ii) *the fourth-column difference $\Delta_m = \Pi_{p,4}^{(q)}(m) - \Pi_{p,4}^{(0)}(m)$ is independent of $m \geq 5$;*
- (iii) *for one integer λ , the recurrence is exactly*

$$\boxed{\begin{aligned} P_{p,n}^{[4,\lambda]}(u) &= (1 + \ell_n u) P_{p,n-1}^{[4,\lambda]}(u) + p(n-1) \ell_{n-1} u^2 P_{p,n-2}^{[4,\lambda]}(u) \\ &\quad - p(n-2)(n-1) u^3 P_{p,n-3}^{[4,\lambda]}(u) \\ &\quad + \lambda p(n-3)(n-2)(n-1) u^4 P_{p,n-4}^{[4,\lambda]}(u) \end{aligned}} \pmod{u^5}, \quad (8)$$

with $\lambda \in \mathbb{Z}$. In that case

$$\Delta_m = \frac{\lambda p}{4} \quad (m \geq 5).$$

The member $\lambda = 0$ has no new connected size-four source, whereas every $\lambda \neq 0$ has a genuine connected quadruple contribution.

Proof. The equivalence of (i) and (iii) is the definition of the normalized source: $\eta_n \equiv \lambda$ is exactly $q_{p,n} = \lambda p(n-1)_3$. If (i) holds, Theorem 3.7 has boundary value $\Delta_5 = \lambda p/4$, and the constant sequence $\Delta_m = \lambda p/4$ satisfies (5); first-order uniqueness gives (ii) and the displayed constant. Conversely, if $\Delta_m \equiv c$, then (7) gives $\eta_m = 4c/p$ for every $m \geq 5$, while (6) gives the same value for $m = 4$. Hence (ii) implies (i), completing the equivalence. $\square$

Definition 4.3 (Primitive same-sign CCR-4). The choice

$$\lambda = 1$$

is called the *primitive same-sign fourth polarization*. It is the smallest positive integral extension compatible with the sign convention for the lower Hessenberg interaction bands introduced in Section 5.

Remark 4.4 (What $\lambda = 1$ means). Definition 4.3 is a normalization choice. It gives one concrete CCR-4 recurrence, but it is not asserted that the original DPDA uniquely forces $\lambda = 1$. The theorem-level object is the full λ -family and the exact way in which λ is visible.

5 Four-band Hessenberg and operator realization

The extension parameter has a direct linear-algebraic realization.

Definition 5.1 (Fourth-band Hessenberg matrix). Let $H_{p,\lambda}$ be the infinite lower-Hessenberg matrix with

$$\begin{aligned} (H_{p,\lambda})_{i,i} &= \ell_i, & (H_{p,\lambda})_{i,i+1} &= 1, \\ (H_{p,\lambda})_{i+1,i} &= -pi\ell_i, & (H_{p,\lambda})_{i+2,i} &= -pi(i+1), \\ (H_{p,\lambda})_{i+3,i} &= -\lambda pi(i+1)(i+2), \end{aligned}$$

and all entries below the third subdiagonal equal to zero. Write $H_{p,\lambda,n}$ for its $n \times n$ leading principal truncation and

$$D_{p,n}^{(\lambda)}(u) = \det(I_n + uH_{p,\lambda,n}).$$

Theorem 5.2 (Four-band Hessenberg realization). *The determinants $D_{p,n}^{(\lambda)}$ satisfy recurrence (8) exactly. Consequently*

$$P_{p,n}^{[4,\lambda]}(u) \equiv D_{p,n}^{(\lambda)}(u) \pmod{u^5}.$$

Proof. Expand $\det(I_n + uH_{p,\lambda,n})$ by the last row. The diagonal term gives

$$(1 + \ell_n u) D_{p,n-1}^{(\lambda)}.$$

The first subdiagonal entry closes with one superdiagonal entry and gives

$$+p(n-1)\ell_{n-1}u^2 D_{p,n-2}^{(\lambda)}.$$

The second subdiagonal closes with two consecutive superdiagonal entries and gives

$$-p(n-2)(n-1)u^3 D_{p,n-3}^{(\lambda)}.$$

The third subdiagonal closes with three consecutive superdiagonal entries. Its cofactor sign reverses the negative matrix entry, producing

$$+\lambda p(n-3)(n-2)(n-1)u^4 D_{p,n-4}^{(\lambda)}.$$

No deeper band is present. This is exactly (8). □

Use the exponential basis

$$e_i(x) = \frac{x^{i-1}}{(i-1)!}, \quad i \geq 1.$$

Let $D = \partial_x$, let $Xf = xf$, set $N = XD$, and define

$$L_p = I + (p+1)N.$$

Then

$$De_i = e_{i-1}, \quad Xe_i = ie_{i+1}, \quad X^k e_i = i(i+1) \cdots (i+k-1)e_{i+k},$$

and $L_p e_i = \ell_i e_i$.

Corollary 5.3 (Operator form). *The matrix $H_{p,\lambda}$ is the matrix, in the basis (e_i) , of*

$$\boxed{\mathcal{H}_{p,\lambda} = D + L_p - pXL_p - pX^2 - \lambda pX^3.} \tag{9}$$

Equivalently,

$$\boxed{\mathcal{H}_{p,\lambda} = \left[1 + (p+1)x - p(p+1)x^2 \right] \partial_x + 1 - px - px^2 - \lambda px^3.} \tag{10}$$

Proof. The actions of D , L_p , XL_p , X^2 , and X^3 on e_i reproduce respectively the superdiagonal, diagonal, first subdiagonal, second subdiagonal, and third subdiagonal of $H_{p,\lambda}$. Expanding $L_p = I + (p+1)XD$ gives (10). $\square$

Banded Hessenberg determinants and their finite-term recurrences are classical; see, for example, [1]. The point here is the exact identification of the fourth chronological parameter with the third subdiagonal and the X^3 term.

6 Fourth-column transport and exact identifiability

For the fourth-order family define

$$C_{n,k}^{(p,\lambda)} = [u^k]P_{p,n}^{[4,\lambda]}(u), \quad 0 \leq k \leq 4,$$

and, for $m \geq k+1$,

$$\Pi_{p,k}^{(\lambda)}(m) = \frac{C_{m-1,k}^{(p,\lambda)}}{(m-1)_k}.$$

Because the λ -term starts in degree four, Theorem 3.6 implies

$$\Pi_{p,k}^{(\lambda)} = \Pi_{p,k} \quad (k = 1, 2, 3).$$

Theorem 6.1 (Fourth-column transport). *For $m \geq 5$,*

$$\boxed{m\Pi_{p,4}^{(\lambda)}(m+1) = (m-4)\Pi_{p,4}^{(\lambda)}(m) + \ell_m\Pi_{p,3}(m) + p\ell_{m-1}\Pi_{p,2}(m-1) - p\Pi_{p,1}(m-2) + \lambda p.} \quad (11)$$

The last term is exactly the normalized connected quadruple source.

Proof. Extract u^4 from (8) and put $n = m$. Divide by

$$(m)_4 = m(m-1)(m-2)(m-3).$$

The four source terms simplify using

$$\begin{aligned} \frac{(m-1)_3}{(m)_4} &= \frac{1}{m}, & \frac{(m-1)(m-2)_2}{(m)_4} &= \frac{1}{m}, \\ \frac{(m-2)(m-1)(m-3)_1}{(m)_4} &= \frac{1}{m}, & \frac{(m-3)(m-2)(m-1)}{(m)_4} &= \frac{1}{m}. \end{aligned}$$

Multiplying by m gives (11). $\square$

The key payoff is that the new connected source does not distort the shape of the fourth normalized column; it translates it.

Theorem 6.2 (Connected-quadruple translation law). *For every $\lambda, \mu \in \mathbb{Z}$ and every $m \geq 5$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m) = \frac{p(\lambda - \mu)}{4}.} \quad (12)$$

Equivalently,

$$\boxed{C_{m-1,4}^{(p,\lambda)} - C_{m-1,4}^{(p,\mu)} = \frac{p(\lambda - \mu)}{4}(m-1)_4.} \quad (13)$$

Proof. Subtract the two instances of (11). With

$$\Delta_m = \Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(\mu)}(m),$$

one obtains

$$m\Delta_{m+1} = (m-4)\Delta_m + p(\lambda - \mu).$$

The constant sequence

$$\Delta_m = \frac{p(\lambda - \mu)}{4}$$

satisfies this recurrence. At the first admissible row $m = 5$, direct extraction from $P_{p,4}^{[4,\lambda]}$ gives the same difference. Uniqueness of the first-order recurrence proves (12); multiplying by $(m-1)_4$ gives (13). $\square$

Corollary 6.3 (One-row identifiability). *Once the lower-order polarization is fixed, any single normalized fourth-column value at an admissible row identifies λ exactly:*

$$\lambda = \frac{4}{p} \left(\Pi_{p,4}^{(\lambda)}(m) - \Pi_{p,4}^{(0)}(m) \right).$$

At the first row $n = 4$, changing λ by one changes $[u^4]P_{p,4}$ by exactly $6p$.

Thus order four is simultaneously non-identifiable from the entire third jet and exactly identifiable from one fourth-column observation.

Theorem 6.4 (Explicit normalized fourth column). *For $m \geq 5$,*

$$\Pi_{p,4}^{(\lambda)}(m) = \frac{G_{p,\lambda}(m)}{5760}, \quad (14)$$

where

$$\begin{aligned} G_{p,\lambda}(m) = & 15(p+1)^4 m^4 + 30(p+1)^3(3p-1)m^3 \\ & - 5(p+1)^2(287p^2 + 158p - 1)m^2 \\ & + 2(p+1)^2(1645p^2 - 50p + 1)m \\ & + 240p(6\lambda - 5p^3 - 5p^2 + 11p + 5). \end{aligned} \quad (15)$$

In particular,

$$\Pi_{p,4}^{(\lambda)}(5) = \frac{\lambda p + 5p^3 + 15p^2 + 13p + 4}{4}. \quad (16)$$

Proof. Substitute the three explicit polynomials of Theorem 2.2 into the source of (11). The resulting source is polynomial in m of degree four. Direct expansion verifies that (14) satisfies (11); evaluation at $m = 5$ gives (16). The first-order transport equation then gives uniqueness. The λ -dependent part of (15) is $1440\lambda p$, which is exactly the translation $\lambda p/4$ of Theorem 6.2. $\square$

7 The fourth-jet Stirling shadow

The new connected parameter is exact information, but it disappears modulo p .

Theorem 7.1 (Fourth-jet Stirling shadow). *For every prime $p \geq 5$, every $\lambda \in \mathbb{Z}$, and every $n \geq 0$,*

$$P_{p,n}^{[4,\lambda]}(u) \equiv \prod_{j=1}^n (1 + ju) \pmod{(p, u^5)}. \quad (17)$$

Consequently,

$$\boxed{C_{m-1,4}^{(p,\lambda)} \equiv e_4(1, \dots, m-1) \equiv \begin{bmatrix} m \\ m-4 \end{bmatrix} \pmod{p},} \quad (18)$$

where the bracket is the unsigned Stirling number of the first kind.

Proof. Modulo p , one has $\ell_n \equiv n$. Every pair, triple, and quadruple interaction source in (8) contains an explicit factor of p , so the recurrence reduces to

$$P_{p,n}^{[4,\lambda]}(u) \equiv (1 + nu)P_{p,n-1}^{[4,\lambda]}(u) \pmod{(p, u^5)}.$$

Iterating gives (17). The coefficient of u^4 in the product is the fourth elementary symmetric function of $1, \dots, m-1$, which is the near-diagonal unsigned Stirling number shown in (18); see, for example, [2]. $\square$

Over $\mathbb{Q}[m]$ the normalized fourth shadow is

$$\boxed{S_4(m) = \frac{m(15m^3 - 30m^2 + 5m + 2)}{5760},} \quad (19)$$

because

$$e_4(1, \dots, m-1) = (m-1)_4 S_4(m).$$

Corollary 7.2 (Normalized shadow away from $p = 5$). *For every prime $p \geq 7$,*

$$\boxed{\Pi_{p,4}^{(\lambda)}(m) \equiv S_4(m) \pmod{p}}$$

as a polynomial congruence. The congruence is independent of λ .

Proof. Subtract (19) from the exact quartic (14). The resulting rational polynomial has an explicit factor of p . For $p \geq 7$, the denominator

$$5760 = 2^7 \cdot 3^2 \cdot 5$$

is invertible modulo p , so reduction is legitimate. $\square$

Remark 7.3 ($p = 5$ denominator resonance). The raw coefficient congruence (18) remains valid at $p = 5$ because it is obtained before any division. However, 5760 is divisible by 5, so one must not obtain a normalized polynomial congruence by cancelling $(m-1)_4$ or reducing (19) coefficientwise modulo 5. This is a normalization-denominator issue, not a failure of the fourth-jet Stirling shadow.

Corollary 7.4 (Shadow blindness versus exact identifiability). *All stationary p -divisible fourth polarizations have the same mod- p fourth jet, but distinct parameters have distinct exact fourth columns. Thus the Stirling shadow forgets λ , while Theorem 6.3 recovers it from one exact row.*

8 Computational regression

All results above are symbolic. Exact computer algebra was used only as an independent regression check of signs, indexing, and polynomial expansions.

For the concrete primes

$$p \in \{5, 7, 11, 13\}$$

and parameters

$$\lambda \in \{-2, -1, 0, 1, 2\},$$

the recurrence (8) was compared with the truncated determinant $\det(I_n + uH_{p,\lambda,n})$ for every $0 \leq n \leq 8$. All 180 instances agreed through u^4 . A separate check used

$$p \in \{5, 7, 11, 13, 17\}, \quad \lambda \in \{-3, -1, 0, 1, 4\}, \quad 5 \leq m \leq 12,$$

and compared the normalized fourth coefficient extracted from the recurrence with (14); extending the row range to $5 \leq m \leq 13$ gives 225 exact checks, all of which agreed. The same computation confirmed that the coefficients through u^3 are independent of λ .

The new inverse theorem Theorem 3.7 was stress-tested separately. For each

$$p \in \{5, 7, 11, 13, 17\},$$

we generated 20 independent integer source sequences $\eta_4, \dots, \eta_{12} \in [-4, 4]$, propagated the fourth-column difference by the exact transport recurrence, and then reconstructed every source entry from (7), including the boundary value (6). Across 800 boundary/transport reconstruction checks there was no discrepancy.

These checks are not used as proofs. They are regression evidence for the symbolic determinant, source-reconstruction, transport, translation, and quartic calculations.

9 Related work and positioning

The vocabulary of “connected” fourth-order information has several classical neighbors, and the distinctions matter.

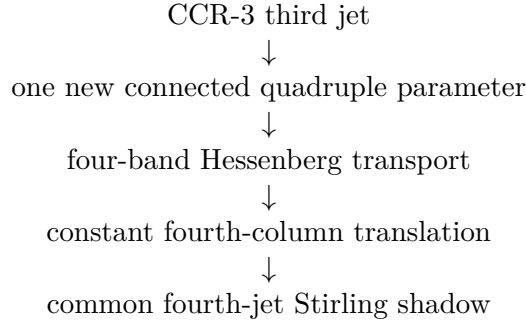
Cumulants and Möbius inversion. Speed formulates cumulants through the Möbius function of the partition lattice [5], building on the general theory of Möbius functions developed by Rota [4]. The familiar vanishing of mixed cumulants across independent families explains why cumulants isolate connected dependence. Our Theorem 3.5 uses this only as an analogy. The CCR terminal-block separation is a deterministic recurrence decomposition on an ordered path, not a probabilistic cumulant identity.

Carries. Carry processes and carry matrices have a substantial classical literature; Holte’s “amazing matrix” is a well-known example [3]. The present refinement does not claim a new general carry process. Its data are the specific ℓ_n -weighted chronological recurrence and its connected low-order deformation.

Hessenberg recurrences. Finite-term recurrences represented by banded lower-Hessenberg matrices are standard. For example, Coussement and Van Assche relate banded Hessenberg matrices to higher-order recurrences for multiple orthogonal polynomials and recover the recurrence by determinant expansion [1]. Our matrix calculation is elementary in that general context. The family-specific statement is that the new chronological parameter occupies exactly the third subdiagonal and becomes the $-\lambda p X^3$ operator term.

Stirling numbers. Unsigned Stirling numbers of the first kind and their elementary-symmetric realization are classical; see [2]. The fourth-jet shadow is not a new Stirling identity. What matters here is that every connected interaction deformation is p -divisible, so all values of the new exact parameter collapse to the same Stirling shadow.

Position of the contribution. A targeted search located these adjacent theories but no direct theorem giving the complete chain

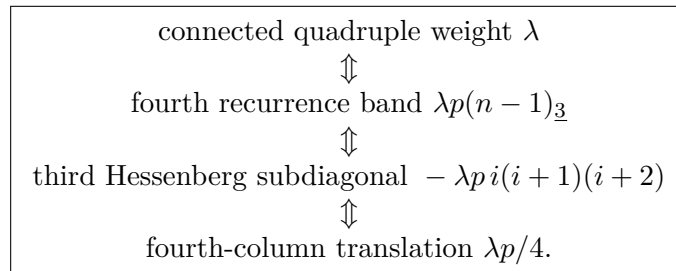


Accordingly, the novelty claim is restricted to this family-specific extension and identifiability package. No priority claim is made for cumulants, carry matrices, Hessenberg recurrences, or Stirling-number formulas themselves.

10 Synthesis

The fourth-order problem has a different answer from a naive extrapolation of CCR-3. Path geometry does identify the new motif: four connected chronological events must form one consecutive quadruple. But geometry and the complete third jet do not identify its numerical arithmetic weight. The new weight is genuinely additional data.

Within the stationary p -divisible class, that data is exactly one integer λ . It has four mutually consistent descriptions:



The mod- p fourth jet then supplies a complementary negative statement: it cannot see λ at all. Every extension has the same free Stirling shadow. Exact arithmetic therefore distinguishes fourth-order chronological polarizations that the universal shadow identifies.

The primitive choice $\lambda = 1$ provides a concrete fourth-order recurrence when one wants a single normalized model. The more important theorem is the classification itself: order four is the first point at which the chosen CCR-3 arithmetic polarization acquires a new independent connected coordinate.

Statements and Declarations

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Data and Code Availability. The paper is theoretical and uses no empirical dataset. All definitions, recurrences, constructions, and symbolic proofs needed for the mathematical results are contained in the manuscript. Exact computer-algebra checks described in Section 8 are independent consistency tests and are not required to establish the proofs.

Ethics Approval and Consent to Participate. Not applicable.

Consent for Publication. Not applicable.

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